MATHEMATICS / INDEPENDENT STUDY

# MM5020

# Mathematics III - Abstract Algebra

Complete solutions

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**How to use this volume** E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give a decisive idea, a complete argument, and criteria for self-assessment.

# 1 Unit 1 · Groups and cyclic structure

### MM5020-U1.01 · E · 15 min
Learning objective: U1.1 Group axioms

Show that the even integers form a group under addition, and explain why they do not form a group under multiplication.

**Decisive idea.** The multiplicative identity would have to be one.

Sums and negatives of even integers are even; addition is associative and has identity zero. Thus they form an additive group. A multiplicative identity inside this set would satisfy $e \cdot 2 = 2$, forcing $e = 1$, which is not even. Hence multiplication does not give a group.

**Check your argument.** Check closure, associativity, identity, and inverse; a single failed multiplicative axiom suffices.

### MM5020-U1.02 · E · 15 min
Learning objective: U1.1 Cancellation

Solve $axb = c$ in an arbitrary group and prove uniqueness.

**Decisive idea.** Multiply first by $a^{-1}$ on the left and then by $b^{-1}$ on the right.

The solution is $a^{-1}cb^{-1}$. Substitution gives $a(a^{-1}cb^{-1})b = c$. Any solution must have this value after the indicated multiplications, proving uniqueness.

**Check your argument.** Keep the factor order; do not commute arbitrary elements.

### MM5020-U1.03 · E · 15 min
Learning objective: U1.2 Element orders

Find the order and generated subgroup of $[9]$ in $\mathbb{Z}/15\mathbb{Z}$.

**Decisive idea.** The order is $\frac{15}{\gcd(15,9)}$.

The order is five. The successive multiples are $[0]$, $[9]$, $[3]$, $[12]$, $[6]$, then $[0]$. These five elements form the generated subgroup.

**Check your argument.** Give both the order and the actual residue classes.

### MM5020-U1.04 · E · 15 min
Learning objective: U1.2 Subgroup test

Prove that the intersection of two subgroups of a group is a subgroup.

**Decisive idea.** Apply the one-step subgroup test in both subgroups.

Let $H, K \leq G$. Their intersection contains $e$. For $a, b \in H \cap K$, both subgroups contain $ab^{-1}$, so it lies in the intersection. The subgroup test proves the claim.

**Check your argument.** Explain nonemptiness as well as closure under $ab^{-1}$.

### MM5020-U1.05 · C · 30 min
Learning objective: U1.1 Structural identities

Prove that a group in which $x^2 = e$ for every $x$ is abelian.

**Decisive idea.** Compare $(ab)^{-1}$ with $b^{-1}a^{-1}$.

For any $a, b$, the hypothesis gives $a^{-1} = a$, $b^{-1} = b$, and $(ab)^{-1} = ab$. The inverse-of-product identity gives $ab = ba$. Thus the group is abelian.

**Check your argument.** Apply the hypothesis to the product as well as the individual factors.

### MM5020-U1.06 · C · 30 min
Learning objective: U1.2 Cyclic subgroups

List all subgroups of $\mathbb{Z}/18\mathbb{Z}$ and their orders.

**Decisive idea.** For each divisor $d$ of eighteen, use $\langle [d] \rangle$.

The subgroups are generated by $[1], [2], [3], [6], [9], [18]$, with orders $18, 9, 6, 3, 2, 1$ respectively. Every subgroup is cyclic; choosing its least positive representative shows its generator divides eighteen, so the list is exhaustive.

**Check your argument.** Justify completeness, not only the six constructions.

### MM5020-U1.07 · C · 30 min
Learning objective: U1.2 Generator arithmetic

Let $a$ have order $n$. Prove $\langle a^r \rangle = \langle a^{\gcd(n,r)} \rangle$.

**Decisive idea.** If $d = un + vr$, then $a^d = (a^r)^v$.

Write $d = \gcd(n, r)$. Since $d$ divides $r$, $a^r$ is a power of $a^d$, giving one inclusion. Bézout gives $d = un + vr$, so $a^d = a^{un}(a^r)^v = (a^r)^v$, giving the reverse inclusion.

**Check your argument.** Prove both inclusions and use $a^n = e$.

### MM5020-U1.08 · C · 30 min
Learning objective: U1.2 Subgroup structure

Prove that the union of two subgroups $H, K$ is a subgroup exactly when one contains the other.

**Decisive idea.** For $h \in H \setminus K$ and $k \in K \setminus H$, test $hk$.

If one subgroup contains the other, their union is that subgroup. Otherwise choose $h \in H \setminus K$ and $k \in K \setminus H$. If $hk \in H$, then $k = h^{-1}(hk) \in H$, a contradiction. If $hk \in K$, then $h = (hk)k^{-1} \in K$, also a contradiction. Thus the union is not closed.

**Check your argument.** Use the correct multiplication side in each contradiction.

### MM5020-U1.09 · C · 30 min
Learning objective: U1.1 Noncommutative examples

Let $r$ rotate a square by a quarter turn and $s$ reflect it in a vertical axis. Show $srs = r^{-1}$ and use this to prove the symmetry group is nonabelian.

**Decisive idea.** Reflection reverses orientation, hence reverses the direction of rotation.

Label vertices cyclically by residues modulo four so that $r(i) = i + 1$ and write the reflection as $s(i) = c - i$ modulo four for a fixed residue $c$. Then $srs(i) = c - (c - i + 1) = i - 1 = r^{-1}(i)$. If $r, s$ commuted, this would give $r = r^{-1}$, contradicting the order four of $r$.

**Check your argument.** Give a precise labeling or coordinate calculation.

### MM5020-U1.10 · A · 50 min
Learning objective: U1.2 Automorphisms

Classify all automorphisms of the additive group $\mathbb{Z}/n\mathbb{Z}$.

**Decisive idea.** The image must generate the group.

Every endomorphism is $f_{k([r])} = [kr]$, since additivity determines all values from $[1]$. This is well defined modulo $n$. Its image is the subgroup generated by $[k]$, of order $\frac{n}{\gcd(n,k)}$. Thus it is bijective exactly when $\gcd(n, k) = 1$. The automorphisms are precisely these multiplication maps; composition corresponds to multiplication of unit residues modulo $n$.

**Check your argument.** Prove that every map is accounted for and determine the exact invertibility condition.

### MM5020-U1.11 · A · 50 min
Learning objective: U1.2 Products of cyclic groups

Prove that $C_m \times C_n$ is cyclic exactly when $\gcd(m, n) = 1$.

**Decisive idea.** A pair of generators has order $\operatorname{lcm}(m, n)$.

For any pair $(x, y)$, its order divides $\operatorname{lcm}(m, n)$ because that exponent kills both coordinates. If the gcd exceeds one, this least common multiple is smaller than $mn$, so no element generates the product. If the gcd is one, a pair of generators has order exactly $\operatorname{lcm}(m, n) = mn$: an exponent killing both is divisible by both $m, n$. Thus this pair generates.

**Check your argument.** Prove impossibility in the noncoprime case for every pair, not just one candidate.

### MM5020-U1.12 · A · 50 min
Learning objective: U1.1 Axioms and finiteness

Let $M$ be a finite nonempty set with an associative operation and a two-sided identity. Assume left and right cancellation. Prove that $M$ is a group.

**Decisive idea.** On a finite set it is surjective, so left and right inverses exist.

For $a \in M$, both maps $x \mapsto ax$ and $x \mapsto xa$ are injective by cancellation and hence surjective. Choose $b, c$ with $ab = e$ and $ca = e$. Associativity gives $c = c(ab) = (ca)b = b$, so $b$ is a two-sided inverse. Every element has an inverse and the other group axioms were assumed.

**Check your argument.** Identify precisely where finiteness and associativity enter.

# 2 Unit 2 · Cosets and permutations

### MM5020-U2.01 · E · 15 min
Learning objective: U2.1 Cosets

List the cosets of $H = \{[0], [4], [8]\}$ in $\mathbb{Z}/12\mathbb{Z}$.

**Decisive idea.** There are $\frac{12}{3} = 4$ cosets.

They are $\{0, 4, 8\}$, $\{1, 5, 9\}$, $\{2, 6, 10\}$, and $\{3, 7, 11\}$, with entries understood modulo twelve. They are disjoint and cover all twelve residues.

**Check your argument.** List each coset once and account for all elements.

### MM5020-U2.02 · E · 15 min
Learning objective: U2.1 Lagrange

What element orders are possible in a group of order twenty-one according to Lagrange's theorem? Does this alone guarantee an element of order twenty-one?

**Decisive idea.** A necessary condition need not be sufficient.

The possible orders permitted by Lagrange are $1, 3, 7, 21$. Lagrange gives no existence conclusion for order twenty-one; it only says that an element's order, if it occurs, must divide twenty-one.

**Check your argument.** Distinguish an allowed order from a guaranteed order.

### MM5020-U2.03 · E · 15 min
Learning objective: U2.2 Permutation composition

Compute $(123)(23)$ and its sign, composing right to left.

**Decisive idea.** One maps to two, two maps to one, and three is fixed.

The product is $(12)$. Its sign is negative. Independently, the three-cycle is even and the transposition is odd, so the product is odd.

**Check your argument.** State the composition convention and verify images.

### MM5020-U2.04 · E · 15 min
Learning objective: U2.2 Cycle order

Find the order and sign of $(1234)(567)$ in $S_7$.

**Decisive idea.** Take the least common multiple and multiply their signs.

The order is $\operatorname{lcm}(4, 3) = 12$. The four-cycle has sign $(-1)^3 = -1$ and the three-cycle sign $(-1)^2 = 1$, so the product has sign $-1$.

**Check your argument.** Do not add cycle lengths when computing order.

### MM5020-U2.05 · C · 30 min
Learning objective: U2.1 Coset criteria

Prove $aH = bH$ if and only if $a^{-1}b \in H$.

**Decisive idea.** Write $b = ah$.

If the cosets agree, $b \in aH$, so $b = ah$ and $a^{-1}b = h \in H$. Conversely, if $b = ah$ with $h \in H$, then $bH = ahH = aH$ because multiplication by $h$ permutes $H$.

**Check your argument.** Give both implications; either inverse orientation is valid if used consistently.

### MM5020-U2.06 · C · 30 min
Learning objective: U2.1 Prime order

Prove that a group of prime order has no proper nontrivial subgroup and is cyclic.

**Decisive idea.** The subgroup generated by a nonidentity element has order greater than one.

Any subgroup has order dividing the prime $p$, hence order one or $p$. For $g \neq e$, $\langle g \rangle$ is not the trivial subgroup, so it has order $p$ and equals the whole group.

**Check your argument.** Explain the existence of a nonidentity element and why its generated subgroup is nontrivial.

### MM5020-U2.07 · C · 30 min
Learning objective: U2.2 Powers of cycles

Describe the disjoint cycles of $(123456)^2$ and $(123456)^3$, and compute their orders.

**Decisive idea.** The cycles are the orbits of adding two or three.

The square is $(135)(246)$ and has order three. The cube is $(14)(25)(36)$ and has order two. These agree with $\frac{6}{\gcd(6,2)}$ and $\frac{6}{\gcd(6,3)}$.

**Check your argument.** Show the actual cycles, not only the order formula.

### MM5020-U2.08 · C · 30 min
Learning objective: U2.2 Counting permutations

How many three-cycles are in $S_5$? How many products of two disjoint transpositions?

**Decisive idea.** A three-element support has two cycles; a four-element support has three pairings.

There are $\binom{5}{3} \cdot 2 = 20$ three-cycles. For double transpositions choose the omitted fixed point in five ways, then one of three pairings of the remaining four points, giving fifteen.

**Check your argument.** Avoid counting rotations of a cycle or orders of disjoint factors as different permutations.

**MM5020-U2.09 · C · 30 min**
Learning objective: U2.2 Alternating groups
Show every three-cycle is even and every even permutation is a product of three-cycles.
**Decisive idea.** A pair sharing a letter is a three-cycle; a disjoint pair can be split using an intermediate transposition.
A three-cycle is a product of two transpositions, so is even. Pair transpositions in an even factorization. Equal pairs cancel; pairs sharing one letter form a three-cycle or its inverse. For disjoint transpositions, $(ab)(cd) = [(ab)(bc)][(bc)(cd)]$, since the middle transpositions cancel. Each bracket is a three-cycle. Thus every pair can be replaced by three-cycles.
**Check your argument.** Handle equal, overlapping, and disjoint pairs.

**MM5020-U2.10 · A · 50 min**
Learning objective: U2.2 Minimal permutation degree
Find the least $n$ for which $S_n$ has an element of order fifteen.
**Decisive idea.** Compare one fifteen-cycle with disjoint cycles of lengths three and five.
Disjoint cycles of lengths three and five give order fifteen on eight letters. If fewer than eight letters are used, no cycle can have length fifteen. To make the least common multiple divisible by both three and five, there must therefore be distinct cycles with lengths divisible by three and five; these use at least eight letters. Hence the least degree is eight.
**Check your argument.** Prove the lower bound, not merely exhibit an example.

**MM5020-U2.11 · A · 50 min**
Learning objective: U2.1 Indices in a tower
For finite $G$ and $K \leq H \leq G$, prove $[G : K] = [G : H][H : K]$ by constructing coset representatives.
**Decisive idea.** Products of representatives give representatives for $G/K$.
Let $g_i$ represent $G/H$ and $h_j$ represent $H/K$. Every $g$ is $g_i h$ and then $g_i h_j k$, so the products represent all cosets modulo $K$. If $g_i h_j K = g_{i'} h_{j'} K$, then the elements lie in the same coset modulo $H$, forcing $i = i'$. Cancelling $g_i$ forces $h_j K = h_{j'} K$ and hence $j = j'$. Count the products.
**Check your argument.** Establish both coverage and uniqueness.

**MM5020-U2.12 · A · 50 min**
Learning objective: U2.2 Conjugation and cycles
Prove $\tau(a_1 \ldots a_r)\tau^{-1} = (\tau(a_1) \ldots \tau(a_r))$. Deduce that conjugate permutations have the same cycle lengths, and conversely.
**Decisive idea.** For the converse, choose a bijection matching corresponding cycle positions.
The conjugate sends $\tau(a_i)$ to $\tau(a_{i+1})$ cyclically and fixes points outside the displayed support. Thus conjugation relabels each cycle and preserves lengths. Conversely, if two permutations have matching cycle lengths, define $\tau$ by matching the ordered entries of corresponding cycles, including fixed points. The displayed formula makes the conjugate equal to the target permutation.
**Check your argument.** Include fixed points and verify that the matching defines a bijection.

# 3 Unit 3 · Homomorphisms and quotients

### MM5020-U3.01 · E · 15 min
Learning objective: U3.1 Kernels

Find the kernel and image of $f : \mathbb{Z} \to \mathbb{Z}/8\mathbb{Z}$, $f(n) = [3n]$.

**Decisive idea.** Its inverse is also three.

The equation $3n = 0$ modulo eight is equivalent to $n = 0$ modulo eight, so the kernel is $8\mathbb{Z}$. Every residue $[a]$ is $f(3a)$ because $9a = a$ modulo eight; the image is the whole target.

**Check your argument.** Give explicit reasons for both the kernel and surjectivity.

### MM5020-U3.02 · E · 15 min
Learning objective: U3.1 Normality

Show the kernel of the determinant map $\mathrm{GL}_2(\mathbb{R}) \to \mathbb{R}^\times$ is normal.

**Decisive idea.** The kernel consists of matrices of determinant one.

$\det(AB) = \det(A)\det(B)$, so determinant is a group homomorphism. Its kernel is $\mathrm{SL}_2(\mathbb{R})$. Every homomorphism kernel is normal; alternatively $\det(ABA^{-1}) = \det(B) = 1$ for $B$ in the kernel.

**Check your argument.** Identify the kernel and cite or prove the applicable normality criterion.

### MM5020-U3.03 · E · 15 min
Learning objective: U3.2 Quotient size

Let $|G| = 60$ and $N \lhd G$ have order twelve. Determine $|G/N|$ and its group structure.

**Decisive idea.** A group of order five is cyclic.

The quotient has $\frac{60}{12} = 5$ elements. Since five is prime, every nonidentity quotient element generates it, so $G/N \cong C_5$.

**Check your argument.** Use normality for the quotient group and Lagrange for its size.

### MM5020-U3.04 · E · 15 min
Learning objective: U3.1 Homomorphisms

Show a homomorphism $f : G \to H$ satisfies $f(g^n) = f(g)^n$ for every integer $n$.

**Decisive idea.** Use preservation of inverses for negative exponents.

Induction and multiplication preservation prove the identity for positive $n$. For $n = 0$, both sides are the identity because $f(e) = e$. If $n = -m < 0$, then $f(g^n) = f\big((g^{-1})^m\big) = f(g^{-1})^m = f(g)^{-m}$.

**Check your argument.** Do not omit the negative or zero cases.

**MM5020-U3.05 · C · 30 min**
Learning objective: U3.2 First isomorphism theorem
Identify $\mathbb{R}^\times/\{1,-1\}$ using a natural homomorphism.
**Decisive idea.** Its image is the multiplicative group of positive real numbers.
The map $x \mapsto |x|$ is a surjective multiplicative homomorphism onto $\mathbb{R}_{>0}$ with kernel $\{1,-1\}$. The first isomorphism theorem gives the quotient isomorphism $[x] \mapsto |x|$.
**Check your argument.** Specify the target operation and compute the kernel.

**MM5020-U3.06 · C · 30 min**
Learning objective: U3.1 Normality
Prove the center $Z(G)$ is a normal subgroup.
**Decisive idea.** Central elements are fixed by every conjugation.
If $a, b$ commute with every $g$, then so does $ab^{-1}$, since $gab^{-1} = agb^{-1} = ab^{-1}g$. The identity is central, so this is a subgroup. For $z \in Z(G)$, $gzg^{-1} = z$, proving normality.
**Check your argument.** Distinguish closure from the separate conjugation argument.

**MM5020-U3.07 · C · 30 min**
Learning objective: U3.2 Quotient maps
For positive integers $m, n$, when is $[a]_m \mapsto [a]_n$ well defined? When it is, find its kernel.
**Decisive idea.** The necessary and sufficient condition is $n|m$.
Necessity follows because $[m]_m = [0]_m$ forces $[m]_n = 0$. If $n|m$, a difference divisible by $m$ is divisible by $n$, proving sufficiency. The kernel is the set of multiples of $n$ modulo $m$, namely $\{[0], [n], \ldots, \left[\left(\frac{m}{n} - 1\right)n\right]\}$, of order $\frac{m}{n}$.
**Check your argument.** Prove necessity and sufficiency, then list the kernel in the correct source group.

**MM5020-U3.08 · C · 30 min**
Learning objective: U3.2 Correspondence
List all subgroups of $\mathbb{Z}/12\mathbb{Z}$ by using subgroups of $\mathbb{Z}$ containing $12\mathbb{Z}$.
**Decisive idea.** Containment $12\mathbb{Z} \subseteq d\mathbb{Z}$ means $d|12$.
The possible $d$ are $1, 2, 3, 4, 6, 12$. Their images are $d\mathbb{Z}/12\mathbb{Z}$, generated by $[d]$, with orders $\frac{12}{d}$. The correspondence theorem proves there are no others.
**Check your argument.** Pay attention to the reversal between divisibility and ideal/subgroup containment.

**MM5020-U3.09 · C · 30 min**
Learning objective: U3.1 Cyclic quotient by center
Prove that if $G/Z(G)$ is cyclic, then $G$ is abelian.
**Decisive idea.** Write $x = g^r z$ and $y = g^s w$ with $z, w$ central.
Choose $gZ(G)$ generating the quotient. Then write $x = g^r z$ and $y = g^s w$ for integers $r, s$ and $z, w \in Z(G)$, since their cosets are powers of the generating coset. Since $z, w$ commute with everything, $xy = g^{r+s} zw = g^{s+r} wz = yx$. Hence $G$ is abelian.
**Check your argument.** Explain why the expressions cover all elements and why the central factors may move.

**MM5020-U3.10 · A · 50 min**
Learning objective: U3.2 Universal property
Prove that a homomorphism $f : G \to H$ factors through $G/N$ exactly when $N \subseteq \ker f$, and prove uniqueness of the factor.
**Decisive idea.** For sufficiency define $\overline{f}(gN) = f(g)$ and check representatives.
If $f = \overline{f}\pi$, then $n \in N$ maps to $\overline{f}(N) = e$, so $N \subseteq \ker f$. Conversely the containment makes equal cosets have equal images, because $b^{-1}a \in N$ implies $f(a) = f(b)$. The candidate respects products by the homomorphism law. Every coset is $\pi(g)$, forcing any factor's value there to be $f(g)$, so it is unique.
**Check your argument.** Give both directions and all well-definedness details.

**MM5020-U3.11 · A · 50 min**
Learning objective: U3.2 Abelianization
Let $[G, G]$ be generated by all commutators $aba^{-1}b^{-1}$. Show it is normal, its quotient is abelian, and every map from $G$ to an abelian group factors uniquely through this quotient.
**Decisive idea.** An abelian target kills each commutator.
Conjugating a commutator by $g$ gives the commutator of $gag^{-1}$ and $gbg^{-1}$, so conjugation preserves the generated subgroup. In the quotient each commutator is the identity, precisely the equation making the cosets of $a, b$ commute. If $f$ has abelian target, the image of every commutator is the identity, so $[G, G] \subseteq \ker f$. The quotient universal property gives the unique factorization.
**Check your argument.** Show normality for the generated subgroup, not only for individual generators.

**MM5020-U3.12 · A · 50 min**
Learning objective: U3.1 Failure of converse to Lagrange
Prove that $A_4$ has no subgroup of order six. You may use that three-cycles generate $A_4$.
**Decisive idea.** The quotient would have order two, but every three-cycle must map to its identity.
Such a subgroup would be normal by the index-two theorem, yielding a surjective homomorphism $A_4 \to C_2$. The image of an element of order three has order dividing both three and two, hence is the identity. Since three-cycles generate $A_4$, the homomorphism would be trivial, contradicting surjectivity.
**Check your argument.** Explain why the image order divides the original order and why generators determine the map.

# 4 Unit 4 · Actions and simple groups

**MM5020-U4.01 · E · 15 min**
Learning objective: U4.1 Actions
Verify that $S_n$ acts on $\{1, \ldots, n\}$ by evaluation.
**Decisive idea.** The operation in $S_n$ is function composition.
The identity permutation fixes every element. Also $(\sigma\tau) \cdot i = \sigma(\tau(i)) = \sigma\tau \cdot i$ by the definition of composition. These are exactly the action axioms.
**Check your argument.** Explicitly check both axioms.

**MM5020-U4.02 · E · 15 min**
Learning objective: U4.1 Stabilizers
Find the stabilizer of one in the natural $S_4$ action and its order.
**Decisive idea.** It is isomorphic to $S_3$.
The stabilizer contains exactly the permutations fixing one. Restriction to $\{2, 3, 4\}$ identifies it with $S_3$, so it has six elements. Orbit–stabilizer agrees: $\frac{24}{6} = 4$.
**Check your argument.** Give the subgroup description as well as its size.

**MM5020-U4.03 · E · 15 min**
Learning objective: U4.2 Conjugacy
Compute $(123)(12)(123)^{-1}$.
**Decisive idea.** Send one to two and two to three.
The result is $(23)$, by the formula $\tau(ab)\tau^{-1} = (\tau(a)\tau(b))$.
**Check your argument.** Identify the conjugating permutation and its two relevant images.

**MM5020-U4.04 · E · 15 min**
Learning objective: U4.2 Simplicity
Which nontrivial finite cyclic groups are simple?
**Decisive idea.** A cyclic group of composite order has a proper nontrivial subgroup.
Exactly the cyclic groups of prime order. Prime order permits no proper nontrivial subgroup by Lagrange. For composite order $n = ab$ with $a, b > 1$, the subgroup generated by $g^a$ has order $b$, hence is proper, nontrivial, and normal.
**Check your argument.** Address both prime and composite orders.

**MM5020-U4.05 · C · 30 min**
Learning objective: U4.1 Kernel of an action
Show the kernel of the homomorphism associated with an action is the intersection of all stabilizers.
**Decisive idea.** Write the two membership conditions explicitly.
An element $g$ is in the homomorphism kernel exactly when its associated permutation is the identity, meaning $g \cdot x = x$ for every $x$. This is equivalent to membership in every $G_x$, so the kernel equals $\cap_{x \in X} G_x$. It is normal because it is a kernel.
**Check your argument.** Explain the equivalence point by point.

**MM5020-U4.06 · C · 30 min**
Learning objective: U4.1 Conjugate stabilizers
Prove $G_{h \cdot x} = hG_x h^{-1}$.
**Decisive idea.** The condition is equivalent to $h^{-1}gh$ fixing $x$.
$gh \cdot x = h \cdot x$ holds exactly when $(h^{-1}gh) \cdot x = x$. Thus $g$ stabilizes $h \cdot x$ exactly when $h^{-1}gh \in G_x$, which is equivalent to $g \in hG_x h^{-1}$.
**Check your argument.** Prove equality through an if-and-only-if membership calculation.

**MM5020-U4.07 · C · 30 min**
Learning objective: U4.2 Class equation
Use conjugacy classes to show the center of $S_3$ is trivial.
**Decisive idea.** The class sizes are one, three, and two.
The identity is a singleton class. The three transpositions form one class and the two three-cycles another, by cycle-type conjugacy. Only elements in singleton conjugacy classes are central, so $Z(S_3) = \{e\}$.
**Check your argument.** Explain why a singleton conjugacy class characterizes a central element.

**MM5020-U4.08 · C · 30 min**
Learning objective: U4.2 Prime-power groups
Prove a group of order $p^2$ is abelian.
**Decisive idea.** If its order were $p$, the quotient by the center would be cyclic.
The class equation gives a nontrivial center, whose order is $p$ or $p^2$ by Lagrange. In the latter case the group is abelian. In the former, $G/Z(G)$ has prime order and is cyclic; U3.09 then makes $G$ abelian, forcing the center to be all of $G$ and contradicting its supposed order $p$.
**Check your argument.** Treat both possible center sizes and justify the cyclic-quotient step.

**MM5020-U4.09 · C · 30 min**
Learning objective: U4.1 Orbit counting
Count two-colorings of four labeled square vertices up to rotation, with reflections excluded.
**Decisive idea.** The four rotations fix $16, 2, 4, 2$ colorings respectively.
The identity fixes all $2^4 = 16$. A quarter or three-quarter rotation forces all vertices to share one color, giving two each. A half-turn permits independent colors on two opposite pairs, giving four. The orbit count is $\frac{16+2+4+2}{4} = 6$.
**Check your argument.** Use precisely the rotation group requested and explain each fixed-point count.

**MM5020-U4.10 · A · 50 min**
Learning objective: U4.2 Simple prime-power groups
Classify finite simple groups whose order is a power of a prime.
**Decisive idea.** Simplicity forces the center to be the whole group.
For order $p^n$ with $n \geq 1$, the center is nontrivial by the class equation. It is normal, so simplicity forces $Z(G) = G$. Thus $G$ is abelian. A nontrivial finite abelian simple group is cyclic of prime order, so here the only possibility is $C_p$. Conversely $C_p$ is simple.
**Check your argument.** Prove the converse and exclude all higher prime powers.

**MM5020-U4.11 · A · 50 min**
Learning objective: U4.1 Coset actions
Let $H$ have finite index $n$ in $G$. Show there is a normal subgroup $N \subseteq H$ with $[G : N] \leq n!$.
**Decisive idea.** Use the kernel of the resulting map to $S_n$.
Left multiplication defines an action on the $n$ cosets. Let $N$ be its kernel; it is normal. An element of $N$ fixes the coset $H$, so it belongs to $H$. The first isomorphism theorem identifies $G/N$ with the image subgroup of $S_n$, so its order is at most $n!$. This also proves the quotient is finite even if $G$ is infinite.
**Check your argument.** Do not assume $H$ itself is normal or $G$ finite.

**MM5020-U4.12 · A · 50 min**
Learning objective: U4.2 Faithful actions of simple groups
Prove every nontrivial action of a simple group is faithful. Deduce that $A_5$ cannot act nontrivially on a set of four elements.
**Decisive idea.** A faithful action on four points would embed $A_5$ into $S_4$.
The action kernel is either trivial or the whole simple group. A nontrivial action rules out the latter, so it is faithful. Since $A_5$ is simple of order sixty, a nontrivial four-point action would give an injection into $S_4$, which has only twenty-four elements, impossible.
**Check your argument.** Here “nontrivial action” means at least one group element moves at least one point.

# 5 Unit 5 · Rings and their quotients

### MM5020-U5.01 · E · 15 min
Learning objective: U5.1 Units

Find every unit in $\mathbb{Z}/10\mathbb{Z}$ and its inverse.

**Decisive idea.** Test the residues one, three, seven, and nine.

The units are $[1], [3], [7], [9]$. Their inverses are respectively $[1], [7], [3], [9]$, since $3 \cdot 7 = 21 = 1$ modulo ten and $9^2 = 81 = 1$. The remaining residues share a factor with ten and cannot have a product congruent to one.

**Check your argument.** Show why the list is exhaustive.

### MM5020-U5.02 · E · 15 min
Learning objective: U5.1 Zero divisors

Give nonzero matrices $A, B \in M_2(\mathbb{Q})$ with $AB = 0$.

**Decisive idea.** Take diagonal matrices with complementary single nonzero entries.

Take $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Both are nonzero and their product is the zero matrix.

**Check your argument.** Verify the product explicitly.

### MM5020-U5.03 · E · 15 min
Learning objective: U5.2 Ideals

Show $6\mathbb{Z}$ is an ideal of $\mathbb{Z}$, and describe the identity of its quotient.

**Decisive idea.** Multiplying a multiple of six by any integer preserves that form.

Differences of multiples of six are multiples of six, and $r(6k) = 6(rk)$, so it is an ideal. The quotient identity is $[1] = 1 + 6\mathbb{Z}$, since $[1][a] = [a]$.

**Check your argument.** Do not confuse the zero coset with the multiplicative identity.

### MM5020-U5.04 · E · 15 min
Learning objective: U5.1 Quaternions

Find the inverse of $q = 1 + 2i - j$.

**Decisive idea.** Its squared norm is six.

The conjugate is $1 - 2i + j$ and the norm is $1^2 + 2^2 + (-1)^2 = 6$. Thus $q^{-1} = \frac{1-2i+j}{6}$. Multiplication in either order gives one by the norm identity.

**Check your argument.** Use the squared norm rather than its square root as denominator.

**MM5020-U5.05 · C · 30 min**
Learning objective: U5.2 Quotient criteria
Determine whether $(5)$ and $(6)$ are prime or maximal in $\mathbb{Z}$.
**Decisive idea.** A prime-order residue ring is a field; modulo six there are zero divisors.
$\mathbb{Z}/5\mathbb{Z}$ is a field, so $(5)$ is maximal and prime. Modulo six, $[2][3] = 0$ with both factors nonzero, so $(6)$ is not prime and therefore not maximal.
**Check your argument.** Justify both the positive and negative conclusions using quotient properties.

**MM5020-U5.06 · C · 30 min**
Learning objective: U5.2 Evaluation
Show $\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$.
**Decisive idea.** Divide by $x^2+1$ to identify the kernel.
Evaluation at $i$ is a unital ring homomorphism and is surjective because $a+bi$ is the image of $a+bx$. Dividing a polynomial by $x^2+1$ leaves $a+bx$; its value at $i$ vanishes only if $a=b=0$. Thus the kernel is exactly $(x^2+1)$, and the first isomorphism theorem applies.
**Check your argument.** Prove the kernel equality, not just one containment.

**MM5020-U5.07 · C · 30 min**
Learning objective: U5.1 Characteristic
Prove a nonzero domain has characteristic zero or a prime.
**Decisive idea.** Translate the integer factorization into a zero product in the ring.
If the characteristic is a positive composite $n = ab$ with $1 < a, b < n$, then $(a \cdot 1)(b \cdot 1) = n \cdot 1 = 0$. Minimality of $n$ makes both factors nonzero, contradicting the domain property. Thus any positive characteristic is prime.
**Check your argument.** Use minimality to establish that both factors are nonzero.

**MM5020-U5.08 · C · 30 min**
Learning objective: U5.2 Ideals and units
Prove an ideal contains a unit if and only if it is the whole ring. Deduce that a field has exactly two ideals.
**Decisive idea.** An ideal containing one contains every element.
If $u \in I$ is a unit, absorption gives $u^{-1}u = 1 \in I$, then $r = r \cdot 1 \in I$ for every $r$. Conversely the whole ring contains one, a unit. In a field, every nonzero ideal contains a nonzero element, hence a unit, so the only ideals are zero and the whole field.
**Check your argument.** Explain both implications and specify nonzero ideals in the deduction.

**MM5020-U5.09 · C · 30 min**
Learning objective: U5.2 Noncommutative ideals
Show the matrices with zero second column form a left ideal of $M_2(K)$ but not a two-sided ideal.
**Decisive idea.** Multiply $E_{11}$ on the right by $E_{12}$.
The set is an additive subgroup and left multiplication preserves the zero second column. But $E_{11}$ lies in it, while $E_{11}E_{12} = E_{12}$ does not. Hence it is not closed under right multiplication and is not two-sided.
**Check your argument.** Check the specified side of multiplication carefully.

**MM5020-U5.10 · A · 50 min**
Learning objective: U5.2 Chinese remainder theorem
Prove $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ for coprime positive $m, n$.
**Decisive idea.** The kernel is $mn\mathbb{Z}$; use Bézout to construct any prescribed pair of residues.
Let $f(a) = ([a]_m, [a]_n)$. Its kernel consists of integers divisible by both $m, n$, hence by $mn$ because they are coprime. Choose $um + vn = 1$. For a target $([r]_m, [s]_n)$, the integer $rvn + sum$ maps to it. Thus $f$ is surjective with the stated kernel, and the ring isomorphism theorem proves the claim.
**Check your argument.** Verify both congruences in the surjectivity construction.

**MM5020-U5.11 · A · 50 min**
Learning objective: U5.2 Matrix ring ideals
Prove every two-sided ideal of $M_{n(K)}$, for a field $K$, is zero or the whole ring.
**Decisive idea.** Multiplication by matrix units isolates that entry in any desired position.
If $A \in I$ has nonzero entry $a_{ij}$, then $E_{ri}AE_{js} = a_{ij}E_{rs} \in I$ for every $r, s$. Multiplying by the scalar inverse gives every matrix unit in $I$. Their linear combinations yield every matrix, so $I = M_{n(K)}$. Thus any nonzero ideal is the whole ring.
**Check your argument.** Use two-sided absorption and the invertibility of the nonzero coefficient.

**MM5020-U5.12 · A · 50 min**
Learning objective: U5.1 Division rings
Prove quaternion conjugation reverses products and deduce multiplicativity of $N(q) = q\overline{q}$.
**Decisive idea.** Use $\overline{qr} = \overline{r}\,\overline{q}$ and centrality of real norms.
For the basis $1, i, j, k$, the rule follows from the multiplication table: conjugation negates each imaginary basis element, and reversing two distinct imaginary factors negates their product. Equal factors give the same real value in either order. Real bilinearity extends the identity to all $q, r$. Hence $N(qr) = qr\overline{r}\,\overline{q} = qN(r)\overline{q} = N(r)N(q)$, since $N(r)$ is real and central.
**Check your argument.** Check equal as well as distinct imaginary factors, and justify the central scalar step.

# 6 Unit 6 · Polynomials and factorization

**MM5020-U6.01 · E · 15 min**
Learning objective: U6.1 Division
Divide $x^3 + 2x + 1$ by $x^2 + 1$ over $\mathbb{Q}$.
**Decisive idea.** Subtract $x(x^2 + 1)$.
The quotient is $x$ and remainder $x + 1$, since $x^3 + 2x + 1 = x(x^2 + 1) + (x + 1)$ and the remainder has degree less than two.
**Check your argument.** Give and verify the division identity and degree bound.

**MM5020-U6.02 · E · 15 min**
Learning objective: U6.1 Bézout
Find integers $u, v$ with $30u + 21v = 3$.
**Decisive idea.** Use $3 = 21 - 2(30 - 21)$.
The identity is $3 = -2 \cdot 30 + 3 \cdot 21$, so $u = -2, v = 3$. The Euclidean divisions $30 = 21 + 9$ and $21 = 2 \cdot 9 + 3$ explain the back substitution.
**Check your argument.** Provide the identity, not just the gcd.

**MM5020-U6.03 · E · 15 min**
Learning objective: U6.2 Irreducibility
Prove $x^3 - 5$ is irreducible over $\mathbb{Q}$.
**Decisive idea.** Eisenstein applies at five.
The polynomial is primitive and its leading coefficient is not divisible by five. Every lower coefficient is divisible by five, while the constant term is not divisible by twenty-five. Eisenstein therefore proves irreducibility over $\mathbb{Q}$.
**Check your argument.** Check every hypothesis, including the square condition.

**MM5020-U6.04 · E · 15 min**
Learning objective: U6.1 Roots
Factor $x^2 + 1$ over $\mathbb{F}_5$.
**Decisive idea.** Two squared equals four modulo five.
The roots are two and three, so $x^2 + 1 = (x - 2)(x - 3)$ over $\mathbb{F}_5$. Expanding gives $x^2 - 5x + 6 = x^2 + 1$ modulo five.
**Check your argument.** State the coefficient field in the factorization.

**MM5020-U6.05 · C · 30 min**
Learning objective: U6.1 Modular inverses
Find the inverse of $x+1$ in $\mathbb{Q}[x]/(x^2+1)$.
**Decisive idea.** The product is $x^2-1$, which becomes negative two.
Modulo $x^2+1$, $(x+1)(x-1)=-2$. Therefore the inverse is $\frac{1-x}{2}$. Direct multiplication gives $\frac{1-x^2}{2}=1$ in the quotient.
**Check your argument.** Reduce using the quotient relation and check the final product.

**MM5020-U6.06 · C · 30 min**
Learning objective: U6.2 Reduction test
Prove $x^3+x+1$ is irreducible over $\mathbb{Q}$ by reducing modulo two.
**Decisive idea.** Evaluate at the only two elements of $\mathbb{F}_2$.
The reduction has value one at both zero and one, so has no root in $\mathbb{F}_2$ and is irreducible there. The integer polynomial is primitive and monic, so its degree is preserved on reduction. Gauss's lemma and the reduction criterion prove irreducibility over $\mathbb{Q}$.
**Check your argument.** Explain why the root test suffices for degree three and why reduction applies.

**MM5020-U6.07 · C · 30 min**
Learning objective: U6.2 Failure outside PIDs
Show 2 is irreducible but not prime in $\mathbb{Z}\left[\sqrt{-5}\right]$.
**Decisive idea.** There is no element of norm two, but $\left(1+\sqrt{-5}\right)\left(1-\sqrt{-5}\right)=6$.
Define $N\left(a+b\sqrt{-5}\right)=a^2+5b^2$. Multiplying an element by its conjugate gives this norm; conjugation preserves products in this commutative ring, so the norm is multiplicative. Its value is one exactly on the units $1,-1$. If $2=uv$, multiplicativity of the norm gives $4=N(u)N(v)$. The only possible nonunit factorization would require both norms to be two, impossible from $a^2+5b^2=2$. Thus two is irreducible. It divides six, but neither $1+\sqrt{-5}$ nor $1-\sqrt{-5}$ is divisible by two, because their coefficients are not both even. Hence two is not prime.
**Check your argument.** Explain why norm one means a unit and why divisibility by two forces even coefficients.

**MM5020-U6.08 · C · 30 min**
Learning objective: U6.1 Principal ideals
Show $K[x]/(f)$ is a field if and only if $f$ is irreducible, assuming $f$ is nonconstant.
**Decisive idea.** For irreducible $f$, every polynomial not divisible by $f$ is coprime to it.
If $f$ is irreducible, a nonzero class $[g]$ has $\gcd(f,g)=1$, so a Bézout identity gives its inverse. Conversely, if $f=ab$ with both factors of positive smaller degree, then $[a],[b]$ are nonzero classes whose product is zero, impossible in a field.
**Check your argument.** Prove the factors give nonzero classes by their degrees.

### MM5020-U6.09 · C · 30 min
Learning objective: U6.2 Primitive products

Show the product of two primitive integer polynomials is primitive.

**Decisive idea.** Reduce modulo that prime and use that $\mathbb{F}_{p[x]}$ is a domain.

For any prime $p$, primitiveness means neither factor becomes the zero polynomial modulo $p$. Their product is therefore nonzero in the domain $\mathbb{F}_{p[x]}$. Thus no prime divides every coefficient of the integer product, which is exactly primitiveness.

**Check your argument.** The argument must apply to every prime.

### MM5020-U6.10 · A · 50 min
Learning objective: U6.2 Shifted Eisenstein

For prime $p$, prove $1 + x + ... + x^{p-1}$ is irreducible over $\mathbb{Q}$.

**Decisive idea.** The transformed polynomial is $\frac{(y+1)^p-1}{y}$.

Substitute $x = y + 1$. The polynomial becomes $\frac{(y+1)^p-1}{y}$, a polynomial with coefficients $\binom{p}{k}$ for the powers $y^{k-1}$, $1 \leq k \leq p$. Its leading coefficient is one, all others are divisible by $p$, and its constant term is $p$, not divisible by $p^2$. Eisenstein proves it irreducible. Substitution $x \mapsto y + 1$ is an invertible ring change of variable, with inverse $y \mapsto x - 1$, so it preserves reducibility and the original polynomial is irreducible.

**Check your argument.** Justify both binomial divisibility and invariance under the change of variable.

### MM5020-U6.11 · A · 50 min
Learning objective: U6.1 Polynomial Chinese remainder

If $f, g \in K[x]$ are coprime, prove $K[x]/(fg) \cong K[x]/(f) \times K[x]/(g)$.

**Decisive idea.** Construct a preimage of $([a], [b])$ using $uf + vg = 1$.

The map $h \mapsto \left([h]_f, [h]_g\right)$ is a unital homomorphism. A polynomial divisible by both coprime $f, g$ is divisible by $fg$: from $h = fk$ and $g|h$, primeness of irreducible factors of $g$ gives $g|k$, or multiply a Bézout identity by $k$. Its kernel is therefore $(fg)$. Choose $u, v$ with $uf + vg = 1$. Given $a, b$, the polynomial $avg + buf$ has the prescribed remainders. Surjectivity and the first isomorphism theorem finish.

**Check your argument.** Compute the kernel and explicitly verify both residues.

### MM5020-U6.12 · A · 50 min
Learning objective: U6.2 Nonprincipal ideals

Prove $\mathbb{Z}[x]$ is a UFD but not a PID.

**Decisive idea.** A generator of $(2, x)$ would divide both two and $x$.

Since $\mathbb{Z}$ is a UFD, Gauss's theorem makes $\mathbb{Z}[x]$ a UFD. If $(2, x) = (d)$, then $d|2$, so $d$ is a constant integer. Also $d|x$, forcing $d = 1$ or $-1$. That would make the ideal the whole ring. But evaluation at zero followed by reduction modulo two is a homomorphism killing both generators and not one, so the ideal is proper. Contradiction.

**Check your argument.** Separate the UFD proof from the argument excluding a principal generator.

# 7 Unit 7 · Fields as vector spaces

### MM5020-U7.01 · E · 15 min
Learning objective: U7.1 Fractions

Prove $\frac{a}{b} = 0$ in the fraction field of a domain exactly when $a = 0$.

**Decisive idea.** Use the cross-multiplication definition.

Equality with $\frac{0}{1}$ means $a \cdot 1 = b \cdot 0$, hence $a = 0$. Conversely a zero numerator gives that equality directly.

**Check your argument.** Use the equivalence relation defining fractions.

### MM5020-U7.02 · E · 15 min
Learning objective: U7.2 Minimal polynomials

Find the minimal polynomial of $\sqrt{7}$ over $\mathbb{Q}$ and a basis of its generated field.

**Decisive idea.** It is irreducible by Eisenstein at seven.

The minimal polynomial is $x^2 - 7$, since it is monic, vanishes at $\sqrt{7}$, and is irreducible. Hence the extension has degree two and basis $1, \sqrt{7}$.

**Check your argument.** Verify irreducibility rather than merely exhibit a vanishing polynomial.

### MM5020-U7.03 · E · 15 min
Learning objective: U7.1 Tower law

If $[L : K] = 3$ and $[M : L] = 4$, find $[M : K]$. Can an intermediate field between $K$ and $M$ have degree five over $K$?

**Decisive idea.** An intermediate degree must divide the total degree.

The total degree is twelve. If $K \subseteq F \subseteq M$ had $[F : K] = 5$, the tower law would give $12 = [M : F] \cdot 5$, impossible.

**Check your argument.** State the tower in each application.

### MM5020-U7.04 · E · 15 min
Learning objective: U7.2 Power-basis arithmetic

With $\alpha^3 = 2$, simplify $\alpha^5 + \alpha^4$ in the power basis.

**Decisive idea.** The remaining powers should be below three.

$\alpha^5 = 2\alpha^2$ and $\alpha^4 = 2\alpha$, so the result is $2\alpha^2 + 2\alpha$.

**Check your argument.** Keep coefficients in the base field and powers below the minimal-polynomial degree.

**MM5020-U7.05 · C · 30 min**
Learning objective: U7.2 Inverses
Find $\left(2+\sqrt{3}\right)^{-1}$ in $\mathbb{Q}\left(\sqrt{3}\right)$ and explain why every nonzero $a+b\sqrt{3}$ has an inverse in this field.
**Decisive idea.** The denominator is $a^2-3b^2$.
The requested inverse is $2-\sqrt{3}$ since the product is one. In general the inverse is $\frac{a-b\sqrt{3}}{a^2-3b^2}$. If the denominator were zero with a nonzero numerator, then $b\neq 0$ and $\left(\frac{a}{b}\right)^2=3$, contradicting irrationality of $\sqrt{3}$. Thus the denominator is nonzero.
**Check your argument.** Check the denominator cannot vanish for a nonzero element.

**MM5020-U7.06 · C · 30 min**
Learning objective: U7.2 Algebraicity
Prove every element of a finite extension is algebraic.
**Decisive idea.** If the degree is $d$, the $d+1$ elements $1,\alpha,\ldots,\alpha^d$ are dependent.
Put $d=[L:K]$. For any $\alpha\in L$, the $d+1$ powers $1,\alpha,\ldots,\alpha^d$ are linearly dependent. This supplies coefficients $c_i\in K$, not all zero, with $\sum_{i=0}^{d}c_i\alpha^i=0$. The corresponding nonzero polynomial annihilates $\alpha$, so $\alpha$ is algebraic.
**Check your argument.** Explain why the polynomial is nonzero.

**MM5020-U7.07 · C · 30 min**
Learning objective: U7.2 Change of generator
Find the minimal polynomial of $1+\sqrt{2}$ over $\mathbb{Q}$.
**Decisive idea.** A quadratic with no rational root is irreducible.
If $\alpha=1+\sqrt{2}$, then $(\alpha-1)^2=2$, so $\alpha^2-2\alpha-1=0$. The polynomial $x^2-2x-1$ has roots $1\pm\sqrt{2}$, neither rational, hence is irreducible and is the minimal polynomial.
**Check your argument.** Show irreducibility and monicity.

**MM5020-U7.08 · C · 30 min**
Learning objective: U7.1 Fraction universal property
Explain why the reduction map $\mathbb{Z}\to\mathbb{F}_5$ cannot extend to a unital homomorphism $\mathbb{Q}\to\mathbb{F}_5$.
**Decisive idea.** A homomorphism would send $5\cdot\left(\frac{1}{5}\right)=1$ to zero.
Such an extension would send five to zero. Applying it to $5\left(\frac{1}{5}\right)=1$ yields $0\cdot f\left(\frac{1}{5}\right)=1$, a contradiction in the nonzero field $\mathbb{F}_5$. This shows why the fraction-field extension theorem requires nonzero denominators to have nonzero images.
**Check your argument.** Identify the exact relation whose image is impossible.

**MM5020-U7.09 · C · 30 min**
Learning objective: U7.2 Root adjunction
Construct a field containing a root of $x^3 + x + 1$ over $\mathbb{F}_2$ and list a basis over $\mathbb{F}_2$.
**Decisive idea.** Use the quotient by the polynomial.
At zero and one the polynomial has value one, so as a cubic it is irreducible. Therefore $K = \mathbb{F}_2[x]/(x^3 + x + 1)$ is a field. Writing $\alpha = [x]$, the elements $1, \alpha, \alpha^2$ form a basis and $\alpha^3 = \alpha + 1$.
**Check your argument.** Explain why the quotient is a field and why the powers are a basis.

**MM5020-U7.10 · A · 50 min**
Learning objective: U7.1 Compositum degree
Prove $\left[\mathbb{Q}\left(\sqrt{2}, \sqrt{5}\right) : \mathbb{Q}\right] = 4$ and find a basis.
**Decisive idea.** Square a proposed expression $a + b\sqrt{2}$.
If $\sqrt{5} = a + b\sqrt{2}$ with rational $a, b$, squaring gives $5 = a^2 + 2b^2 + 2ab\sqrt{2}$, so $ab = 0$. If $b = 0$, then $a^2 = 5$; if $a = 0$, then $b^2 = \frac{5}{2}$. Neither is a rational square, as prime exponents in a rational square are even. Thus adjoining $\sqrt{5}$ has degree two over $\mathbb{Q}\left(\sqrt{2}\right)$, and the tower law gives four. Products of the two quadratic bases yield $1, \sqrt{2}, \sqrt{5}, \sqrt{10}$.
**Check your argument.** Exclude both cases and invoke the basis form of the tower law.

**MM5020-U7.11 · A · 50 min**
Learning objective: U7.2 Transitivity of algebraicity
Let $\frac{L}{K}$ be algebraic and let $\alpha$ be algebraic over $L$. Prove $\alpha$ is algebraic over $K$, even when $\frac{L}{K}$ is infinite.
**Decisive idea.** Adjoin those coefficients to $K$ first.
Let $c_0, \ldots, c_d \in L$ be the coefficients of a nonzero polynomial annihilating $\alpha$. The field $F = K(c_0, \ldots, c_d)$ is finite over $K$, since it is generated by finitely many algebraic elements. The element $\alpha$ is algebraic over $F$, so $\frac{F(\alpha)}{F}$ is finite. The tower law makes $\frac{F(\alpha)}{K}$ finite. Every element of a finite extension is algebraic, so $\alpha$ is algebraic over $K$.
**Check your argument.** Do not apply a finite-degree tower law directly to the potentially infinite extension $\frac{L}{K}$.

**MM5020-U7.12 · A · 50 min**
Learning objective: U7.2 A primitive generator
Show $\mathbb{Q}\left(\sqrt{2} + \sqrt{3}\right) = \mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)$ and find the minimal polynomial of the sum.
**Decisive idea.** Since $\left(\sqrt{3} - \sqrt{2}\right)\left(\sqrt{3} + \sqrt{2}\right) = 1$, both roots can be recovered.
Put $\alpha = \sqrt{2} + \sqrt{3}$. Then $\alpha^{-1} = \sqrt{3} - \sqrt{2}$, so $\sqrt{3} = \frac{\alpha + \alpha^{-1}}{2}$ and $\sqrt{2} = \frac{\alpha - \alpha^{-1}}{2}$. This proves equality of the fields. Squaring twice gives $\left(\alpha^2 - 5\right)^2 = 24$, hence $\alpha^4 - 10\alpha^2 + 1 = 0$. The field has degree four by Lecture 13, so the minimal polynomial has degree four and must be this monic quartic.
**Check your argument.** Use the degree to prove minimality; a vanishing quartic alone is insufficient.

# 8 Unit 8 · Finite fields and applications

### MM5020-U8.01 · E · 15 min
Learning objective: U8.1 Field orders

Which of the orders $8, 9, 10, 16, 18, 25$ can occur for a finite field?

**Decisive idea.** Apply the existence theorem as well as the necessary condition.

The possible orders are $8 = 2^3$, $9 = 3^2$, $16 = 2^4$, and $25 = 5^2$. Fields of each exist. Ten and eighteen involve distinct primes and are not prime powers, so neither occurs.

**Check your argument.** State both necessity and existence.

### MM5020-U8.02 · E · 15 min
Learning objective: U8.1 Arithmetic in four elements

In $\mathbb{F}_4 = \mathbb{F}_2[\alpha]$ with $\alpha^2 + \alpha + 1 = 0$, compute $\alpha(\alpha + 1)$ and the inverse of $\alpha + 1$.

**Decisive idea.** Addition has characteristic two.

$\alpha(\alpha + 1) = \alpha^2 + \alpha = (\alpha + 1) + \alpha = 1$. Thus the inverse of $\alpha + 1$ is $\alpha$.

**Check your argument.** Use characteristic two correctly.

### MM5020-U8.03 · E · 15 min
Learning objective: U8.2 Generators

Find the order of two in $\mathbb{F}_7^\times$.

**Decisive idea.** The third power is one.

The powers are $2, 4, 1$, so the order is three. Two is therefore not a generator of the six-element multiplicative group.

**Check your argument.** Check that no earlier positive power is one.

### MM5020-U8.04 · E · 15 min
Learning objective: U8.1 Characteristic and dimension

What are the characteristic and dimension over the prime field of a field of order eighty-one?

**Decisive idea.** Eighty-one equals three to the fourth power.

The characteristic is three and the dimension over $\mathbb{F}_3$ is four, because a $d$-dimensional vector space over $\mathbb{F}_3$ has $3^d$ elements.

**Check your argument.** Distinguish characteristic from total cardinality.

### MM5020-U8.05 · C · 30 min
Learning objective: U8.2 Frobenius

Prove the map $a \mapsto a^p$ is an automorphism of a finite field of characteristic $p$.

**Decisive idea.** Use the binomial theorem and the fact a field has no nonzero nilpotents.

In characteristic $p$, binomial coefficients make $(a+b)^p = a^p + b^p$; multiplication and one are also preserved. If $a^p = b^p$, then $(a-b)^p = 0$, so $a = b$. The field is finite, so injectivity implies surjectivity. Thus the map is an automorphism.

**Check your argument.** State where finiteness is used; Frobenius need not be surjective in an infinite field.

### MM5020-U8.06 · C · 30 min
Learning objective: U8.2 Interpolation

Find the degree-less-than-three polynomial over $\mathbb{F}_5$ with $f(0) = 1$, $f(1) = 0$, $f(2) = 2$.

**Decisive idea.** The first value gives $c = 1$; solve the remaining two equations modulo five.

Write $f(x) = ax^2 + bx + 1$ using the value at zero. The other two conditions give $a + b = 4$ and $4a + 2b = 1$ modulo five. Subtract twice the first equation to get $2a = 3$, hence $a = 4$ and $b = 0$. Thus $f(x) = 4x^2 + 1$. Substitution checks all three values; uniqueness follows from the root bound.

**Check your argument.** Use modular inverses and explain uniqueness.

### MM5020-U8.07 · C · 30 min
Learning objective: U8.1 Polynomial roots

Show every element of $\mathbb{F}_q$ is a root of $x^q - x$, and this polynomial has no repeated root.

**Decisive idea.** Its formal derivative is negative one.

Zero is a root. Every nonzero $a$ belongs to a group of order $q-1$, so $a^{q-1} = 1$ and $a^q = a$. Since $q$ is zero in the field, the derivative is $qx^{q-1} - 1 = -1$, which shares no root with the polynomial, so no root repeats.

**Check your argument.** Handle zero separately and use characteristic correctly in the derivative.

### MM5020-U8.08 · C · 30 min
Learning objective: U8.2 Evaluation codes

Evaluate all degree-less-than-two polynomials at the five elements of $\mathbb{F}_5$. How many codewords result, and what is their minimum distance?

**Decisive idea.** A nonzero difference is a nonzero polynomial of degree at most one.

There are $5^2 = 25$ polynomials, and evaluation is injective because a nonzero degree-at-most-one polynomial cannot vanish at five points. Two distinct codewords differ in at least four coordinates. This bound is attained by the polynomials zero and $x$, whose evaluations agree only at zero. Thus the minimum distance is four.

**Check your argument.** Show the bound is attained, not only that it is a lower bound.

### MM5020-U8.09 · C · 30 min
Learning objective: U8.2 Primitive elements

Find all generators of $\mathbb{F}_7^\times$, given that three is one generator.

**Decisive idea.** Use exponents one and five.

The generators are $3^1 = 3$ and $3^5 = 5$ modulo seven. The order formula shows no other powers generate because their exponents share a nontrivial gcd with six.

**Check your argument.** Explain exhaustiveness using the cyclic-group order formula.

### MM5020-U8.10 · A · 50 min
Learning objective: U8.1 Subfields

Show a field of sixteen elements has a unique subfield of four elements.

**Decisive idea.** Its nonzero roots satisfy $x^3 = 1$ in a cyclic group of order fifteen.

The cyclic multiplicative group of order fifteen has exactly three solutions to $x^3 = 1$: for a generator $g$, the exponents satisfy $15|3k$, giving $k = 0, 5, 10$. Including zero gives four roots of $x^4 - x$. The fourth-power Frobenius identity shows these roots are closed under addition and multiplication, and nonzero inverses also remain roots, so they form a subfield. Any four-element subfield consists entirely of such roots and hence equals this one.

**Check your argument.** Prove existence by closure and uniqueness by the root characterization.

### MM5020-U8.11 · A · 50 min
Learning objective: U8.2 Polynomial functions

Prove every function $\mathbb{F}_q \to \mathbb{F}_q$ is represented by a unique polynomial of degree less than $q$. Is the representing polynomial unique without the degree bound?

**Decisive idea.** The polynomial $x^q - x$ represents the zero function.

Interpolation gives a polynomial of degree less than $q$ with any prescribed function values, and the root bound gives uniqueness in that degree range. Without the bound, adding any multiple of $x^q - x$ changes the formal polynomial without changing any value on the field. Thus uniqueness fails among all polynomials.

**Check your argument.** Distinguish formal polynomials from the functions they induce.

### MM5020-U8.12 · A · 50 min
Learning objective: U8.1 Irreducible quadratics

Count the monic irreducible quadratics over $\mathbb{F}_q$.

**Decisive idea.** Reducible monic quadratics correspond to unordered pairs of roots, allowing repetition.

There are $q^2$ monic quadratics. A reducible one has the form $(x - a)(x - b)$ with $a, b \in \mathbb{F}_q$, and unique factorization identifies it with an unordered pair with repetition. There are $\frac{q(q+1)}{2}$ such pairs. Thus the number of monic irreducible quadratics is $q^2 - \frac{q(q+1)}{2} = \frac{q(q-1)}{2}$.

**Check your argument.** Include repeated roots and justify why no reducible quadratic is missed or counted twice.

# 9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

### MM5020-M1.01 · E · 15 min

Learning objective: U1–U2 Cyclic groups

List the orders of every element of $\mathbb{Z}/8\mathbb{Z}$.

**Decisive idea.** For residue $k$, the order is $\frac{8}{\gcd(8,k)}$.

For $k = 0, 1, 2, 3, 4, 5, 6, 7$, the orders are respectively $1, 8, 4, 8, 2, 8, 4, 8$. The formula follows by asking for the least positive $t$ with $8 | kt$.

**Check your argument.** Explain the formula and treat the zero residue.

### MM5020-M1.02 · E · 15 min

Learning objective: U5 Ring quotients

Determine whether $\mathbb{Z}/15\mathbb{Z}$ is a field and list its units.

**Decisive idea.** The product of three and five is zero.

It is not a field, since nonzero $[3]$, $[5]$ multiply to zero. Its units are $[1]$, $[2]$, $[4]$, $[7]$, $[8]$, $[11]$, $[13]$, $[14]$, exactly the residues coprime to fifteen by Bézout.

**Check your argument.** Justify the full unit list.

### MM5020-M1.03 · C · 30 min

Learning objective: U3–U4 Group structure

Let $|G| = 49$. Prove $G$ is abelian, and show that if some element has order forty-nine then $G$ is cyclic.

**Decisive idea.** The center has order seven or forty-nine.

The center is nontrivial. If it had order seven, the quotient by the center would be cyclic of order seven, forcing $G$ abelian and contradicting that center size. Thus the center is all of $G$. An element of order forty-nine has a generated subgroup of that size, equal to $G$.

**Check your argument.** The cyclicity conclusion is conditional; abelian does not automatically mean cyclic.

### MM5020-M1.04 · C · 30 min
Learning objective: U6–U7 Algebraic extensions

Let $\alpha^3 = 3$ over $\mathbb{Q}$. Prove the extension degree is three and find $(\alpha - 1)^{-1}$.

**Decisive idea.** The product with $\alpha^2 + \alpha + 1$ is two.

$x^3 - 3$ is Eisenstein at three, so it is the minimal polynomial and the degree is three. Since $(\alpha - 1)(\alpha^2 + \alpha + 1) = \alpha^3 - 1 = 2$, the inverse is $\frac{\alpha^2+\alpha+1}{2}$.

**Check your argument.** Prove irreducibility and verify the inverse.

### MM5020-M1.05 · A · 50 min
Learning objective: U3 Universal constructions

For normal subgroups $N, M$ of $G$, show $g \mapsto (gN, gM)$ induces an injective homomorphism $G/(N \cap M) \to G/N \times G/M$. When is it surjective?

**Decisive idea.** Surjectivity is equivalent to $G = NM$.

The paired map is a homomorphism with kernel $N \cap M$, so its induced map is injective by the first isomorphism theorem. If it is surjective, any pair $(aN, M)$ has a lift $g \in M$ with $aN = gN$, so $a \in MN = NM$. Conversely if $G = NM$, for prescribed $aN, bM$ write $b^{-1}a = mn$ with $m \in M, n \in N$. Then $g = bm = an^{-1}$ has both prescribed cosets.

**Check your argument.** Give the kernel and prove both directions of the surjectivity criterion.

### MM5020-M1.06 · A · 50 min
Learning objective: U8 Field structure

Prove that the fixed field of Frobenius $x \mapsto x^p$ in $\mathbb{F}_{p^n}$ is exactly $\mathbb{F}_p$.

**Decisive idea.** The prime field already supplies $p$ roots.

Frobenius fixes the prime field because it fixes one and preserves addition. Thus its fixed set contains $p$ elements. Any fixed element is a root of the degree-$p$ polynomial $x^p - x$, which has at most $p$ roots in a field. Hence there are no others and the fixed field is $\mathbb{F}_p$.

**Check your argument.** Show both containment and the cardinality bound.

# 10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

**MM5020-M2.01 · E · 15 min**
Learning objective: U2 Permutations
Find the order and sign of $(123)(45)(67)$.
**Decisive idea.** Take the least common multiple of three, two, and two.
The order is six. Its sign is $(-1)^2(-1)(-1) = 1$, so it is even.
**Check your argument.** Distinguish order from parity.

**MM5020-M2.02 · E · 15 min**
Learning objective: U7 Minimal polynomials
Find the minimal polynomial of $2 + \sqrt{5}$ over $\mathbb{Q}$.
**Decisive idea.** Use $(x-2)^2 - 5$.
The polynomial is $x^2 - 4x - 1$. It vanishes at the given element and has roots $2 \pm \sqrt{5}$, neither rational, so it is irreducible and minimal.
**Check your argument.** Verify irreducibility.

**MM5020-M2.03 · C · 30 min**
Learning objective: U3 Quotient isomorphisms
Prove $S_4/A_4 \cong C_2$.
**Decisive idea.** Its kernel is $A_4$ and a transposition maps to negative one.
Sign is a homomorphism onto $\{1, -1\}$: the identity maps to one and any transposition to negative one. Its kernel is $A_4$. The first isomorphism theorem gives the quotient isomorphism, and the target is cyclic of order two.
**Check your argument.** Establish surjectivity and the kernel.

### MM5020-M2.04 · C · 30 min
Learning objective: U5–U6 Field construction

Prove $\mathbb{F}_3[x]/(x^2+1)$ is a field and find the inverse of $[x+1]$.

**Decisive idea.** Modulo $x^2+1$, the product is negative two, which equals one in $\mathbb{F}_3$.

The values of $x^2+1$ at zero, one, and two are one, two, and two, so the quadratic is irreducible. The quotient is a field. Since $(x+1)(x-1) = x^2-1 = -2 = 1$ in the quotient, the inverse is $[x-1]$.

**Check your argument.** Use arithmetic in the specified characteristic.

### MM5020-M2.05 · A · 50 min
Learning objective: U4 Actions

A group $G$ of order fifteen acts on a set of seven elements. Prove there is a fixed point.

**Decisive idea.** Only sizes one, three, and five fit inside seven.

Every orbit size divides fifteen and is at most seven, so it is one, three, or five. If no fixed point existed, seven would be a sum of threes and fives. With at most one five, the remaining two is impossible; with no five, seven is not divisible by three. Hence an orbit of size one exists, which is a point fixed by all of $G$.

**Check your argument.** A fixed point means fixed by the entire group, not only one element.

### MM5020-M2.06 · A · 50 min
Learning objective: U7–U8 Degrees and subfields

Prove that $\mathbb{F}_{p^3}$ has no subfield strictly between $\mathbb{F}_p$ and itself.

**Decisive idea.** The tower law makes the intermediate degree divide three.

For an intermediate field $F$, the tower law gives $3 = \left[\mathbb{F}_{p^3} : F\right]\left[F : \mathbb{F}_p\right]$. Both factors are positive integers, so one is one. If $\left[F : \mathbb{F}_p\right] = 1$, then $F = \mathbb{F}_p$; otherwise $\left[\mathbb{F}_{p^3} : F\right] = 1$ and $F$ is the whole field.

**Check your argument.** Explain why degree one forces equality of fields.